SUPERSEDING
MIL-S-19500/293(NAVY)
27 July 1964
(See 6.4)

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, TYPE 1N93A

1. SCOPE

1.1 Description. - This specification covers the detail requirements for a germanium semiconductor diode for use in rectifier circuitry and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Ratings. -

	V_r 60 cycle	V_R	Lead current (see 6.3)		i_f(surge) (1/120 sec)	Barometric pressure
			I_o	T_A		
	v	Vdc	mAdc	°C	a	mmHg
Minimum	---	---	---	---	---	8
Maximum	300	200	175	40	25	---
	200	100	300	50	25	---
	100	50	500	60	25	---

1/ Operating temperature: T_A = -65° to +90°C(see 6.2).

2/ Storage ambient temperature: T_{stg} = -65° to +95°C.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-S-19500 - Semiconductor Devices, General Specification For.

STANDARDS

MILITARY
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

FSC 5960

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3. REQUIREMENTS

3.1 General. - The diodes shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Design, construction and physical dimensions. - The diodes shall be of the design, construction and physical dimensions of JEDEC outline DO-3.

3.3 Performance characteristics. - Performance characteristics shall be as specified in 4.3, 4.4 and 4.5

3.4 Marking. - Diodes shall be marked in accordance with MIL-S-19500, except that country of origin and manufacturer's identification may be omitted from the required marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests. - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.3, 4.4, and 4.5. (Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.1)).

4.2 Quality conformance inspection. - Quality conformance inspection shall consist of the examinations and tests specified in 4.3, 4.4 and 4.5.

4.2.1 Quality conformance inspection information. - When requested by the bureau or agency concerned, one copy of the quality conformance inspection data pertinent to the inspection lot shall accompany the shipments.

4.3 Group A inspection. - Group A inspection shall consist of the examinations and tests shown in table I.

4.4 Group B inspection. - Group B inspection shall consist of the examinations and tests shown in table II.

4.4.1 Shock. - The shock testing apparatus shall be capable of providing shock pulses of the peak acceleration, waveform, and pulse duration specified herein to the body of the device. The acceleration pulse, as determined from the unfiltered output of a transducer with a natural frequency greater than 10,000 cycles per second, shall be a half sine waveform with an allowable distortion not greater than ±20 percent of the peak acceleration specified herein. The pulse duration shall be measured between the points at 10 percent of the peak acceleration during rise and at 10 percent of the peak acceleration during decay. Absolute tolerance of the pulse duration shall be ±40 percent of the duration specified herein.

4.5 Group C inspection. - Group C inspection shall consist of the examinations and tests shown in table III, and shall be performed once every 6 months.

4.6 Destructive tests. - The tests in subgroups 2 and 4 of group B and group C are considered to be destructive.

4.6.1 Terminal strength. - At the conclusion of the testing in subgroup 4 of group B inspection, the devices shall be examined for evidence of mechanical damage.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120.)

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6.2 Operating temperature is measured with a thermocouple imbedded in the cathode lead solder joint. Lead length is 1/2 inch.

6.3 These ratings are based on 1/2 inch anode and cathode lead lengths. Anode and cathode operating and ambient temperatures are assumed equal.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activity:
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Table I - Group A inspection.

Examination or test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 method	Specific conditions			Min.	Max.	
Subgroup I			10				
Visual and mechanical examination	2071	--		--	--	--	--
Subgroup 2			5				
Forward voltage	4011	I_F = 0.5 Adc		V_F	---	0.55	V
Reverse current at peak reverse voltage	4046	I_o = O T_A = 50°C v_r = 300 Vac		i_r	---	1000	uA
Reverse current at peak reverse voltage	4046	I_o = O T_A = -65°C v_r = 300 Vac		i_r	---	200	uA
Reverse current	4016	V_R = 200 Vdc		I_R	---	400	uA

Table II - Group B inspection.

Examination or test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 method	Specific conditions			Min.	Max.	
Subgroup 1			10				
Physical dimensions	2066						
Subgroup 2			10				
Solderability	2026						
Thermal shock (temperature cycling)	1051 Condition B	$T_{(high)}$ = 90°C					
Thermal shock (glass strain)	1056 Condition A						
Moisture resistance	1021						

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Table II - Group B inspection (Cont'd).

Examination or test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 method	Specific conditions			Min.	Max.	
Subgroup 3			10				
Shock	2016	1500G, 0.5msec 3 blows each X, Y, Z (see 4.4.1)					
Vibration fatigue	2046						
Vibration, variable frequency	2056						
Constant acceleration	2006	1500G					
Subgroup 4			10				
Tension	2036 Condition A	5 lbs t = 15 sec					
Lead fatigue	2036 Condition E						
Subgroup 5			10				
Surge current	4066	i_f(surge) = 25a t_p = 1/120 sec v_r = 300 v 60 cycles 10 times, 1/min.					
Subgroup 6			λ= 10				
Operating life	1026	v_r = 300 v 60 cycles I_o = 175 mAdc T_A = 40°C					
Subgroup 7			λ= 10				
Storage life	1031	T_{stg} = 95°C					
End points (for sub-groups 2, 3, 4, 6 and 7:							
Reverse current	4016	V_R = 200 Vdc		I_R	---	400	uA

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Table III - Group C inspection.

Examination or test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 method	Specific conditions			Min.	Max.	
Subgroup 1			10				
Barometric pressure	1001	8 mmHg V_R = 200 Vdc Measured during test		I_R		400	MA
Salt atmosphere	1041						

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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